

Optimum Filter

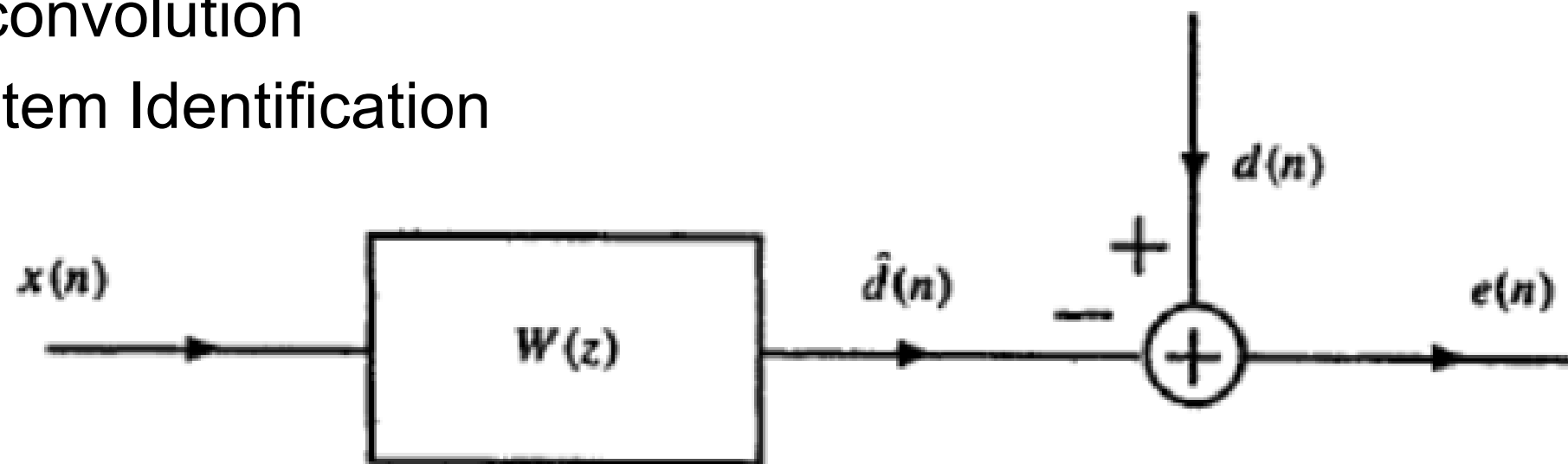
Statistical Signal Processing and Modelling

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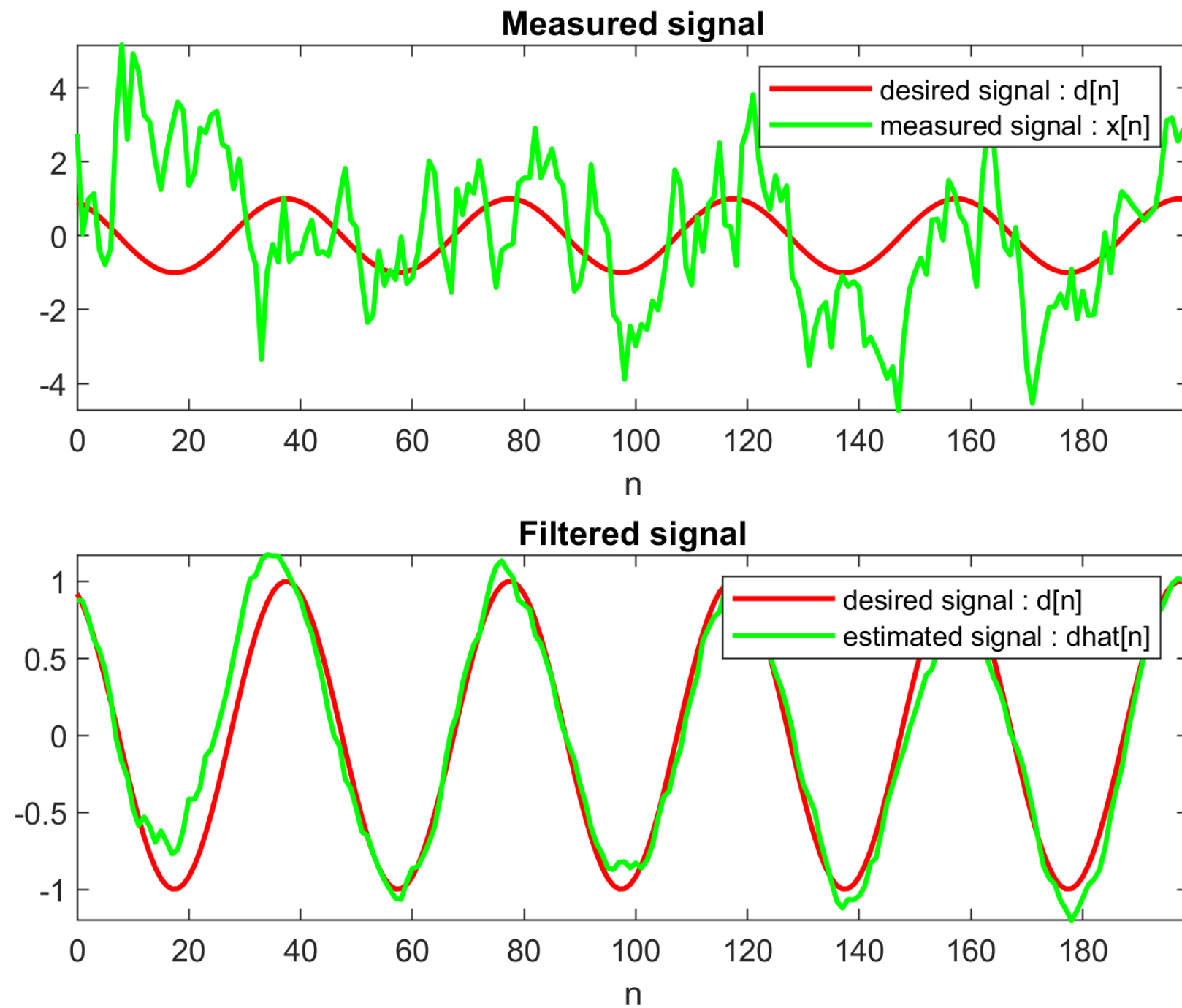
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- FIR Wiener filter
- Kalman filter

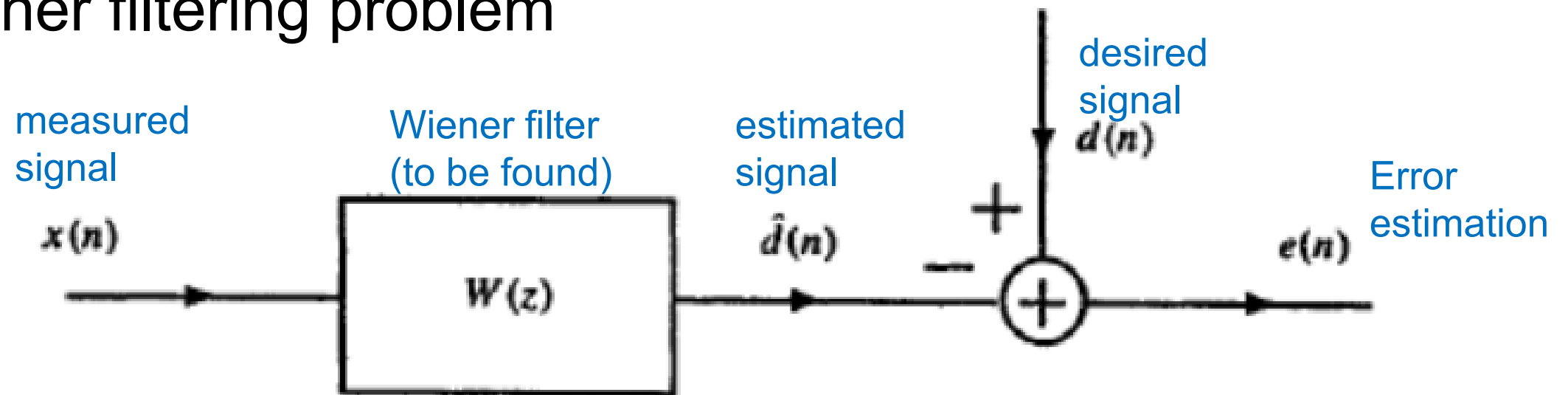
- The goal of an optimal filter is to **estimate one signal from another one**, such that the resulting **mean square error** is as small as possible
- There are many applications to this idea, such as estimating the real signal from a noisy observation in medicine (ECG, MRI, etc.) and other fields:
 - Filtering
 - Smoothing
 - Prediction
 - Deconvolution
 - System Identification



Application : Filtering



- Wiener filtering problem



$$x[n] = d[n] + v[n]$$

measured signal desired signal noise

- Goal : Find an optimal FIR filter $W(z)$

$$W(z) = \sum_{n=0}^{p-1} w[n] \cdot z^{-n} \quad p-1 : \text{filter order}$$

$$\hat{d}[n] = x[n] * w[n] = \sum_{l=0}^{p-1} w[l] \cdot x[n-l]$$

- It is assumed that $x[n]$ and $d[n]$ are **jointly wide-sense stationary (WSS)**
- Goal is to find an optimal FIR filter (coefficients of the filter $w[n]$) which results in the **smallest mean squared error** between the desired signal and the estimated signal

$$\xi = E\{|e[n]|^2\} = E\{|d[n] - \hat{d}[n]|^2\}$$

- To find the smallest mean square error, all derivative must be equal to 0 (for $k=0,1,\dots, p-1$).

$$\frac{\partial \xi}{\partial w^*[k]} = \frac{\partial}{\partial w^*[k]} E\{e[n]e^*[n]\} = E\left\{e[n] \frac{\partial e^*[n]}{\partial w^*[k]}\right\} = 0$$

With

$$e[n] = d[n] - \hat{d}[n] = d[n] - \sum_{l=0}^{p-1} w[l] \cdot x[n-l]$$

- It gives for the derivative term : $\frac{\partial e^*[n]}{\partial w^*[k]} = \dots = -x^*[n-k]$
- And finally:

$$\frac{\partial \xi}{\partial w^*[k]} = E\{e[n]x^*[n-k]\} = 0$$

Leads to the orthogonality principle : data used for the estimation $x[n]$ is orthogonal to the error $e[n]$ and so their dot product is null.

- Based on this orthogonality results, the equations to derive the filter coefficients can be determined:

$$E\{e[n]x^*[n-k]\} = 0 \quad k = 0, 1, \dots, p-1$$

$$E\left\{\left(d[n] - \sum_{l=0}^{p-1} w[l] \cdot x[n-l]\right) x^*[n-k]\right\} = 0$$

$$\underbrace{E\{d[n]x^*[n-k]\}}_{r_{dx}[k]} - \sum_{l=0}^{p-1} w[l] \underbrace{E\{x[n-l]x^*[n-k]\}}_{r_x[k-l]} = 0$$

- It gives the final results

$$\sum_{l=0}^{p-1} w[l] r_x[k-l] = r_{dx}[k]$$

- The previous equation can be written on matrix shape

$$\sum_{l=0}^{p-1} w[l] r_x[k-l] = r_{dx}[k]$$

$$\begin{array}{c}
 k=0 \\
 k=1 \\
 \vdots \\
 k=p-1
 \end{array}
 \begin{array}{c}
 l=0 \quad l=1 \quad \dots \quad l=p-1 \\
 \left[\begin{array}{cccc}
 r_x[0] & r_x[1] & \dots & r_x[p-1] \\
 r_x[1] & r_x[0] & \dots & r_x[p-2] \\
 \vdots & \vdots & \ddots & \vdots \\
 r_x[p-1] & r_x[p-2] & \dots & r_x[0]
 \end{array} \right]
 \left[\begin{array}{c}
 w[0] \\
 w[1] \\
 \vdots \\
 w[p-1]
 \end{array} \right]
 =
 \left[\begin{array}{c}
 r_{dx}[0] \\
 r_{dx}[1] \\
 \vdots \\
 r_{dx}[p-1]
 \end{array} \right]
 \end{array}$$

Toeplitz matrix! R_x
 w
 r_{dx}

- And finally in compact shape also called the **Wiener-Hopf equation**

$$R_x \cdot w = r_{dx}$$

- Finally, the coefficients ($\mathbf{w}[k]$) that minimize the minimum mean square error are obtained from the Wiener-Hopf equations

$$\mathbf{R}_x \cdot \mathbf{w} = \mathbf{r}_{dx}$$

- As the correlation matrix $\mathbf{R}_x$ is Toeplitz, the inversion can be easily computed with Levinson-Durbin

$$\mathbf{w} = \mathbf{R}_x^{-1} \cdot \mathbf{r}_{dx}$$

- The **mean-square error** in the estimate of $d[n]$ may be **estimated** :

$$\xi = E\{|e[n]|^2\} = E \left\{ e[n] \left[d[n] - \sum_{l=0}^{p-1} w[l] \cdot x[n-l] \right]^* \right\} =$$

$$E\{e[n]d^*[n]\} - \sum_{l=0}^{p-1} w^*[l] \cdot \underbrace{E\{e[n]x^*[n-l]\}}_{= 0 \text{ if } w[n] \text{ is the solution of the Wiener-Hopf equations (orthogonality principle)}}$$

- So, the **estimated minimum mean-square error** can be obtained:

$$\xi_{min} = E\{e[n]d^*[n]\} = E \left\{ \left[d[n] - \sum_{l=0}^{p-1} w[l] \cdot x[n-l] \right] d^*[n] \right\} =$$

$$\underbrace{E\{d[n]d^*[n]\}}_{r_d[0]} - \sum_{l=0}^{p-1} w[l] \cdot \underbrace{E\{d^*[n]x^*[n-l]\}}_{r_{dx}^*[l]} = r_d[0] - \mathbf{r}_{dx}^H \mathbf{w}$$

$$\Rightarrow \xi_{min} = r_d[0] - \underbrace{r_{dx}^H \cdot R_x^{-1} \cdot r_{dx}}_{\mathbf{w}}$$

- The Wiener-Hopf equation requires the autocorrelation (R_x) and cross-correlation r_{dx} which are most of the time unknown.

$$R_x \cdot w = r_{dx}$$

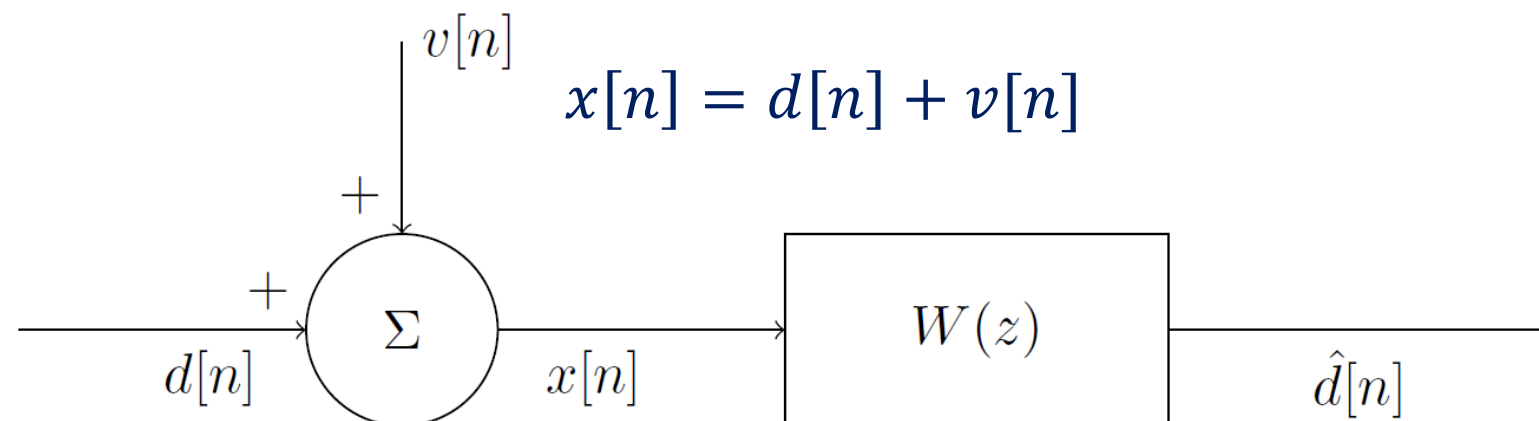
- So they have to be estimated on the signal:

$$\hat{R}_x \cdot w = \hat{r}_{dx}$$

- The quality of the estimation will affect the Wiener filter coefficients

$$\hat{r}_{dx}[k] = \frac{1}{N} \sum_{n=0}^{N-1} d[n] \cdot x^*[n-k]$$

- In filtering, $d[n]$ is observed from a noise corrupted observation



- An usual assumption is that the noise has zero mean and is uncorrelated with the signal

$$E\{d[n]v^*[n - k]\} = 0$$

- **The goal is to find the filter $W(z)$ to estimate $d[n]$**

- The Wiener-Hopf equation can be modified for this application

$$\mathbf{R}_x \cdot \mathbf{w} = \mathbf{r}_{dx}$$

- The cross-correlation by using $x[n] = d[n] + v[n]$ become:

$$r_{dx}[k] = E\{d[n]x^*[n-k]\} = \underbrace{E\{d[n]d^*[n-k]\}}_{r_d[k]} + \underbrace{E\{d[n]v[n-k]\}}_{=0 \text{ because assumption noise is uncorrelated to signal}}$$

$$\Rightarrow r_{dx}[k] = r_d[k]$$

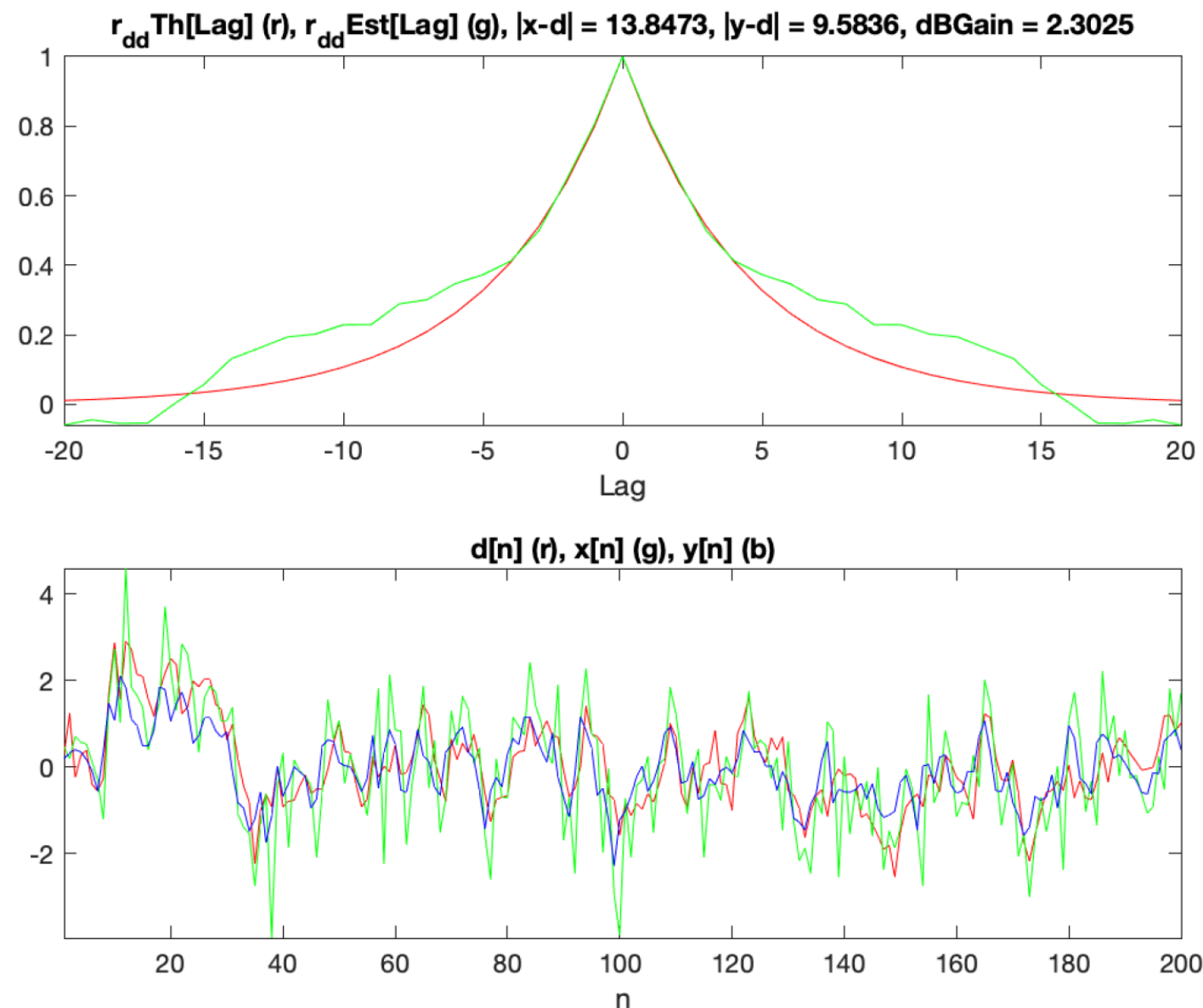
- And the auto-correlation become:

$$r_x[k] = E\{x[n+k]x^*[n]\} = E\{[d[n+k] + v[n+k]][d[n] + v[n]]^*\}$$

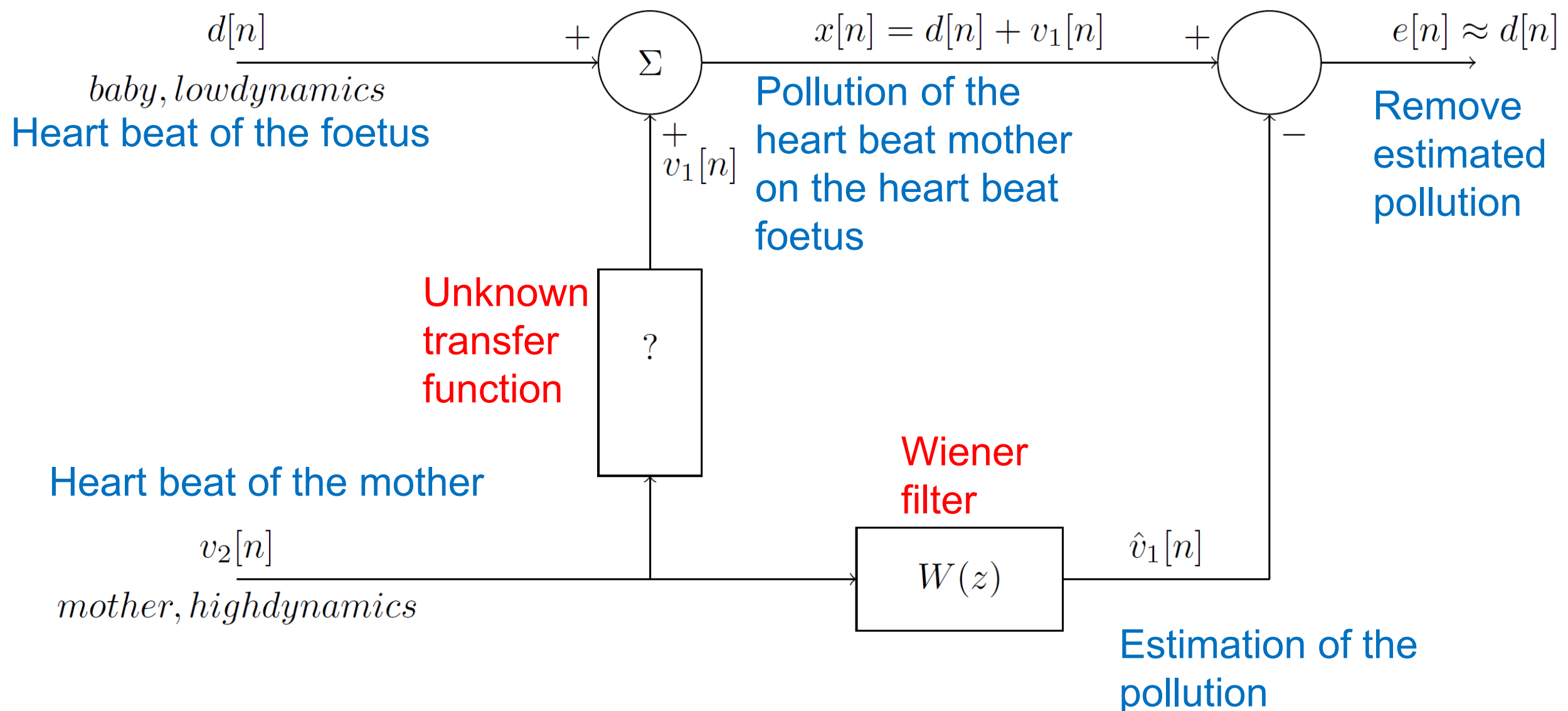
$$\Rightarrow r_x[k] = r_d[k] + r_v[k]$$

- The **Wiener-Hopf equation** become : $[\mathbf{R}_d + \mathbf{R}_v] \cdot \mathbf{w} = \mathbf{r}_d$

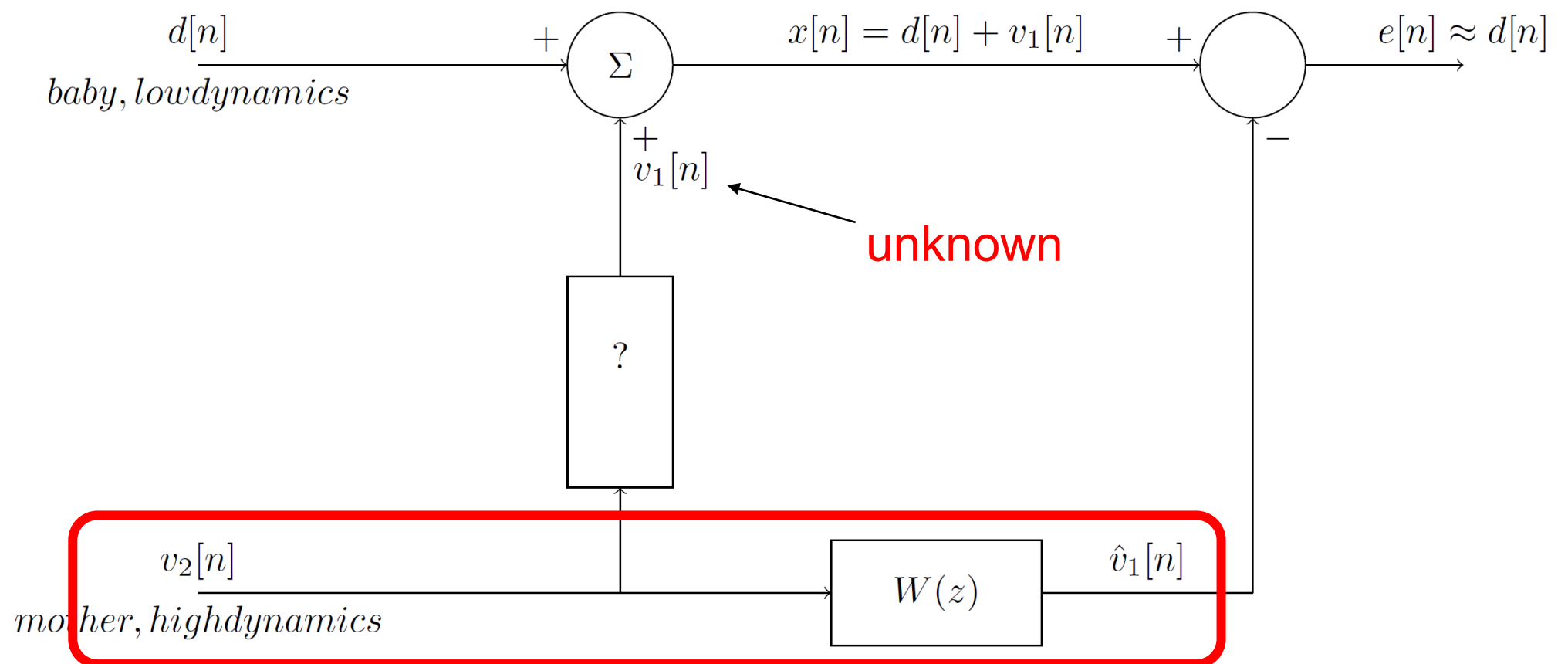
- AR(1) process : $d[n] = 0.8d[n - 1] + w[n]$
- Measured in noisy environment : $x[n] = d[n] + v[n]$
- Wiener filter obtained (p=1) : $\mathbf{w} = (0.405, 0.238)$



- Similar to filtering problem but, this time, an auxiliary noise $v_2[n]$ correlated with measurement noise $v_1[n]$ is used to estimate original signal $d[n]$



- Hence the Wiener-Hopf equation can be formulated again



$$\mathbf{R}_x \cdot \mathbf{w} = \mathbf{r}_{dx} \Rightarrow \mathbf{R}_{v_2} \cdot \mathbf{w} = \mathbf{r}_{v_1 v_2}$$

unknown

- The cross-correlation can be expressed as:

$$\begin{aligned} r_{v_1 v_2}[k] &= E\{v_1[n]v_2^*[n-k]\} = E\{[x[n] - d[n]]v_2^*[n-k]\} \\ &= E\{x[n]v_2^*[n-k]\} - \underbrace{E\{d[n]v_2^*[n-k]\}} \end{aligned}$$

= 0 because signal and noise
are uncorrelated

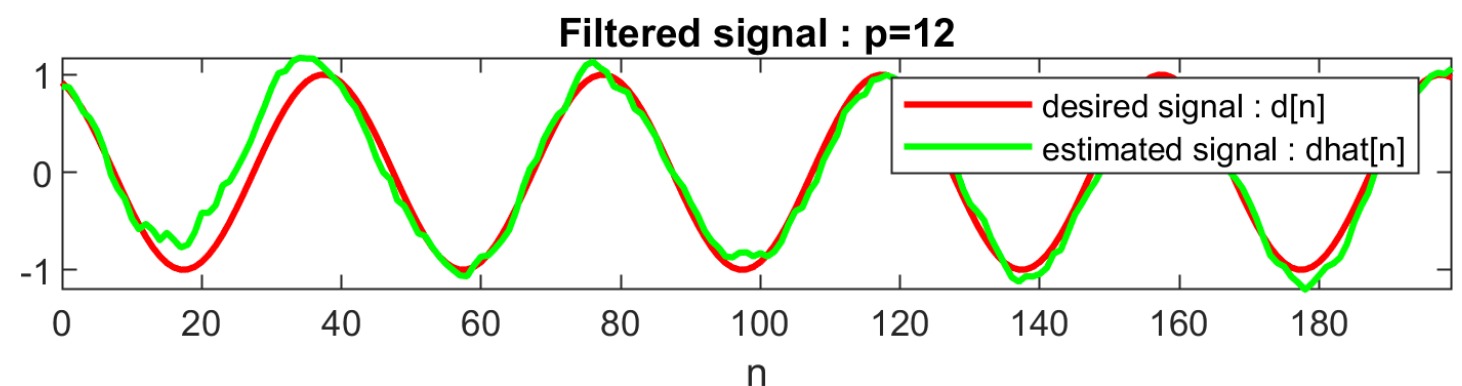
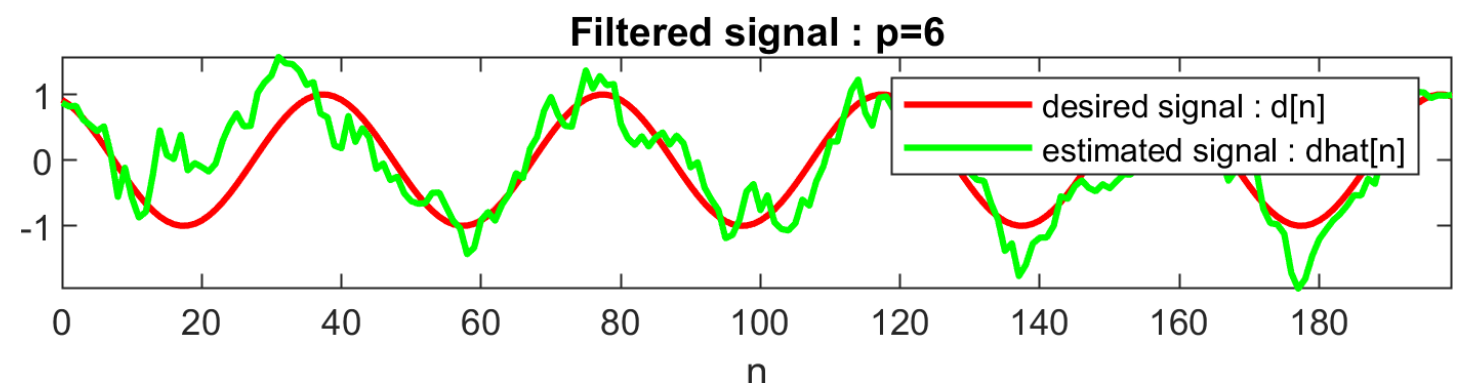
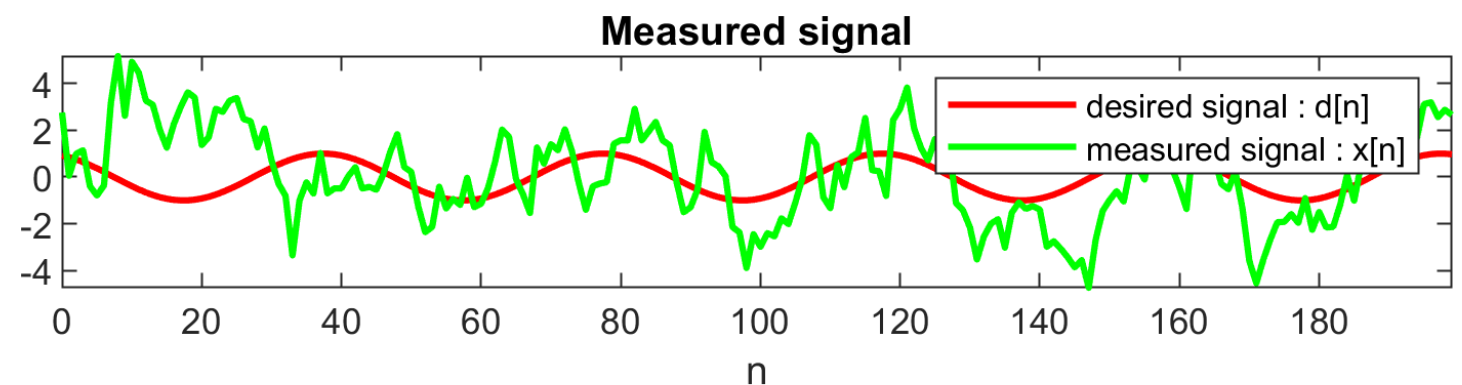
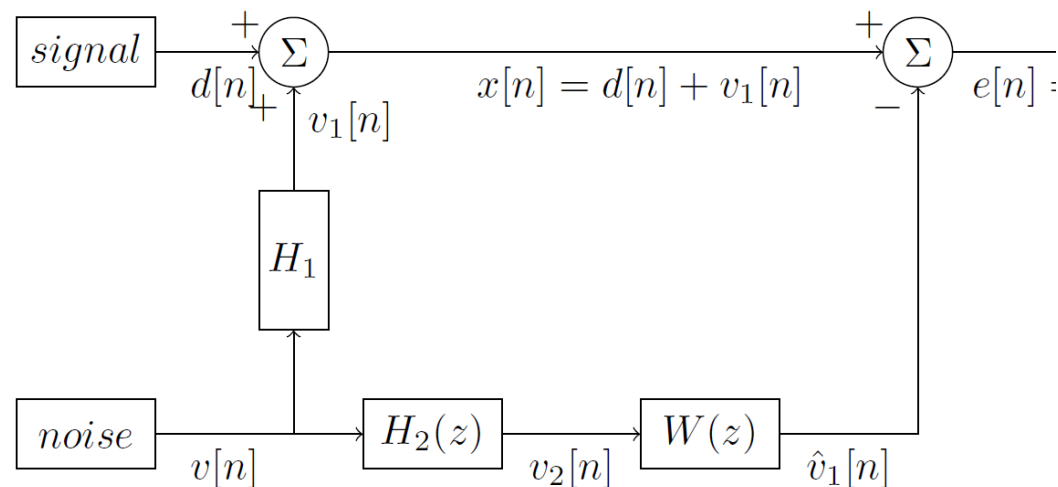
$$\Rightarrow r_{v_1 v_2}[k] = E\{x[n]v_2^*[n-k]\} = r_{xv_2}[k]$$

- Finally, the Wiener-Hopf equations become:

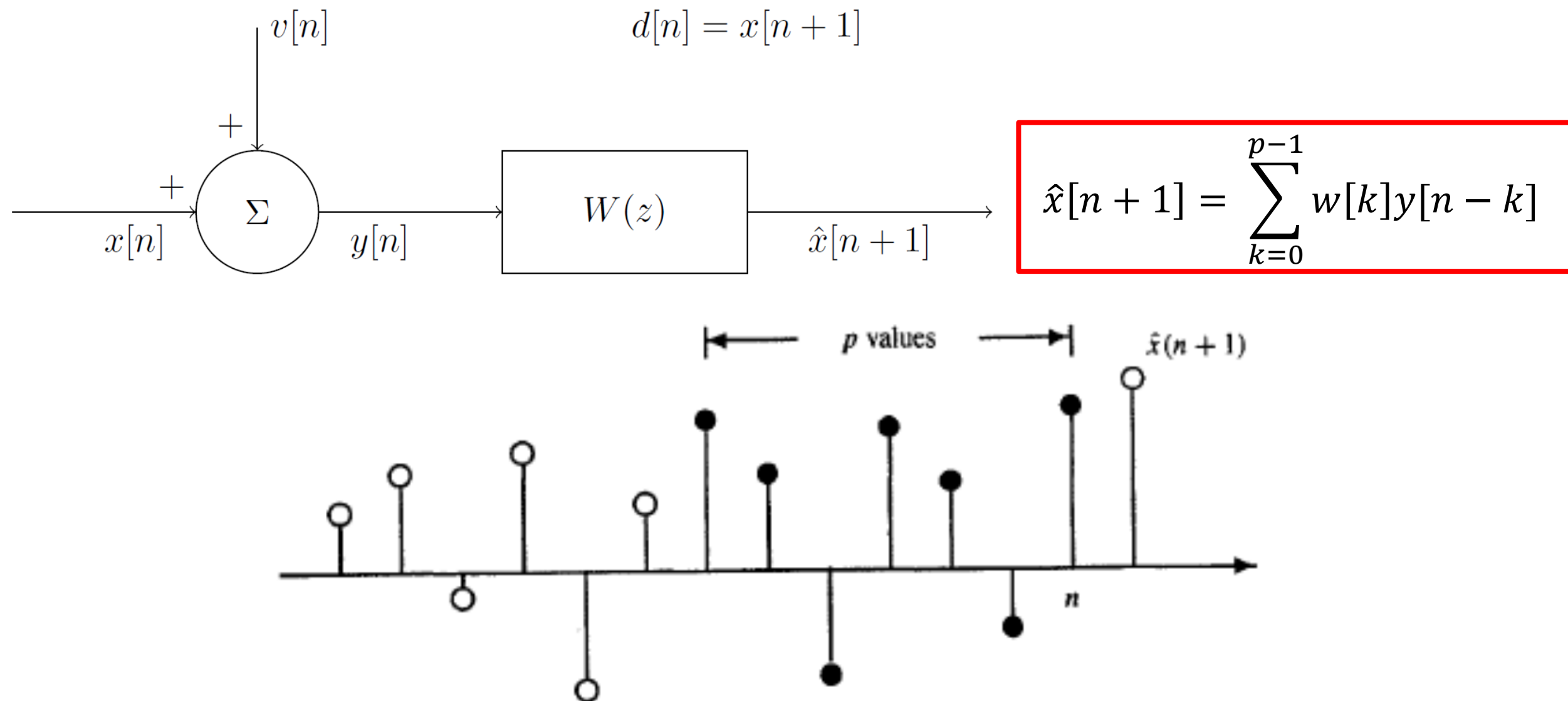
$$\mathbf{R}_{v_2} \cdot \mathbf{w} = \mathbf{r}_{xv_2}$$

Example Noise cancellation (7.2.6)

- Desired signal is : $d[n] = \sin(n\omega_0 + \phi)$
- Measured signal : $x[n] = d[n] + v_1[n]$
- Noise measured : $v_2[n] = -0.6v_2[n - 1] + v[n]$



- The goal is to **predict future samples $x[n+1]$** as a linear combination of past ones:



- The Wiener-Hopf equation can be modified for this application by using $y[n] = x[n] + v[n]$ where $x[n]$ and zero mean white noise $v[n]$ with variance σ_v^2 are uncorrelated
- The cross-correlation by using $d[n] = x[n + \alpha]$ where α is the shifted prediction index become ($\alpha = 1$ for one step)

$$r_{dy}[k] = E\{d[n]y^*[n - k]\} = E\{x[n + \alpha]x^*[n - k]\} + \underbrace{E\{x[n + \alpha]v^*[n - k]\}}_{= 0 \text{ because assumption noise is uncorrelated to signal}}$$

$$= r_x[\alpha + k]$$

- The auto-correlation become

$$r_y[k] = E\{y[n]y^*[n - k]\} = E\{x[n]x^*[n - k]\} + E\{v[n]v^*[n - k]\} = r_x[k] + \underbrace{\sigma_v^2 \delta[k]}_{r_v[k]}$$

- It gives the new Wiener-Hopf equation for multi-step prediction

$$\begin{aligned} \mathbf{R}_y \cdot \mathbf{w} &= \mathbf{r}_{dy} \Rightarrow \\ (\mathbf{R}_x + \sigma_v^2 \mathbf{I}) \cdot \mathbf{w} &= \mathbf{r}_x(\alpha) \end{aligned}$$

- The Wiener-Hopf equation can be expressed with matrix as:

$$(\mathbf{R}_x + \sigma_v^2 \mathbf{I}) \cdot \mathbf{w} = \mathbf{r}_x(\alpha)$$

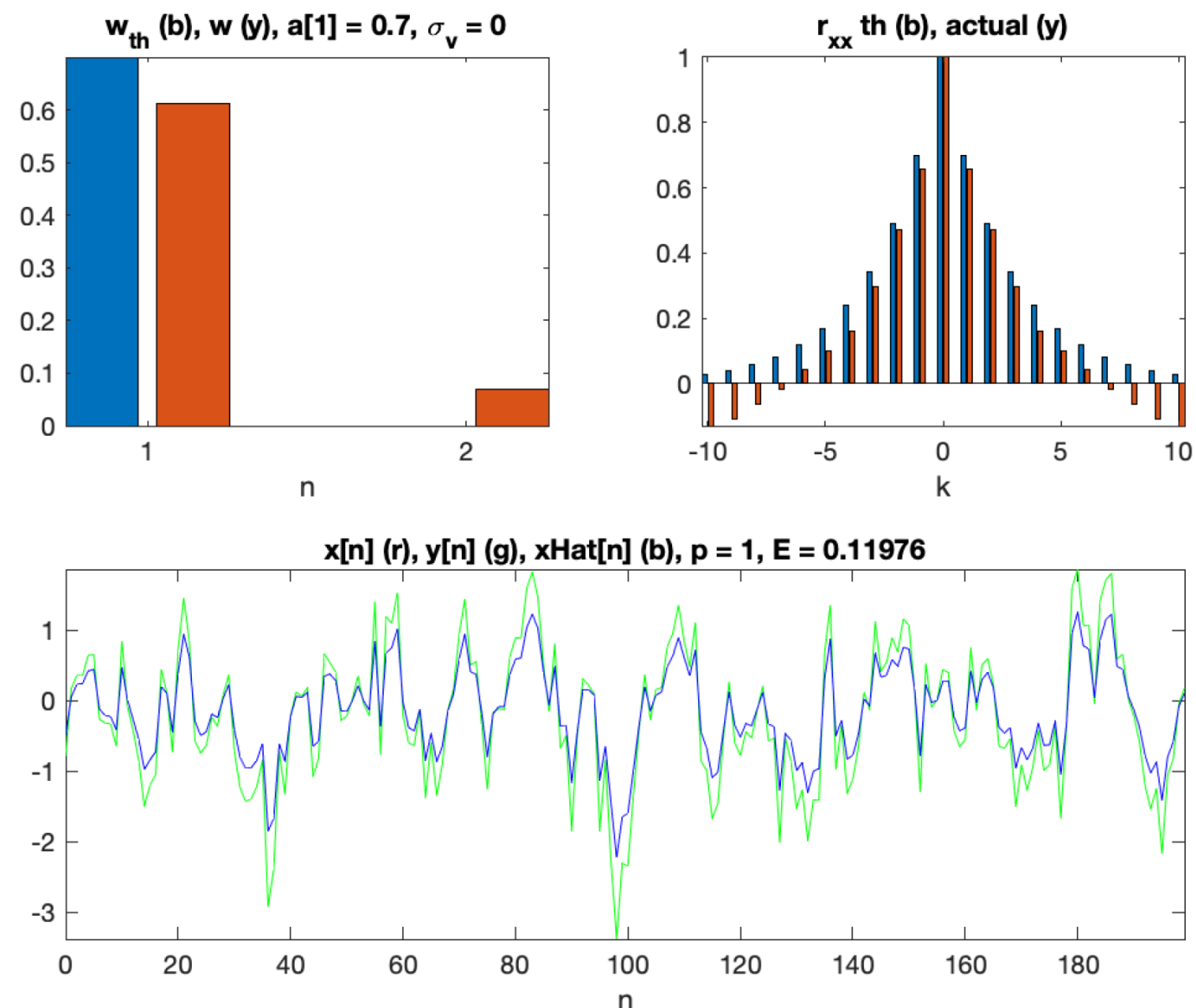
$$\begin{bmatrix} r_x[0] + \sigma_v^2 & r_x[1] & \cdots & r_x[p-1] \\ r_x[1] & r_x[0] + \sigma_v^2 & \cdots & r_x[p-2] \\ \vdots & \vdots & \ddots & \vdots \\ r_x[p-1] & r_x[p-2] & \cdots & r_x[0] + \sigma_v^2 \end{bmatrix} \begin{bmatrix} w[0] \\ w[1] \\ \vdots \\ w[p-1] \end{bmatrix} = \begin{bmatrix} r_x[\alpha] \\ r_x[\alpha+1] \\ \vdots \\ r_x[\alpha+p-1] \end{bmatrix}$$

- If the noise $v[n]$ is null ($\sigma_v^2 = 0$), the Wiener-Hopf is simplified as :

$$\mathbf{R}_x \cdot \mathbf{w} = \mathbf{r}_x(\alpha)$$

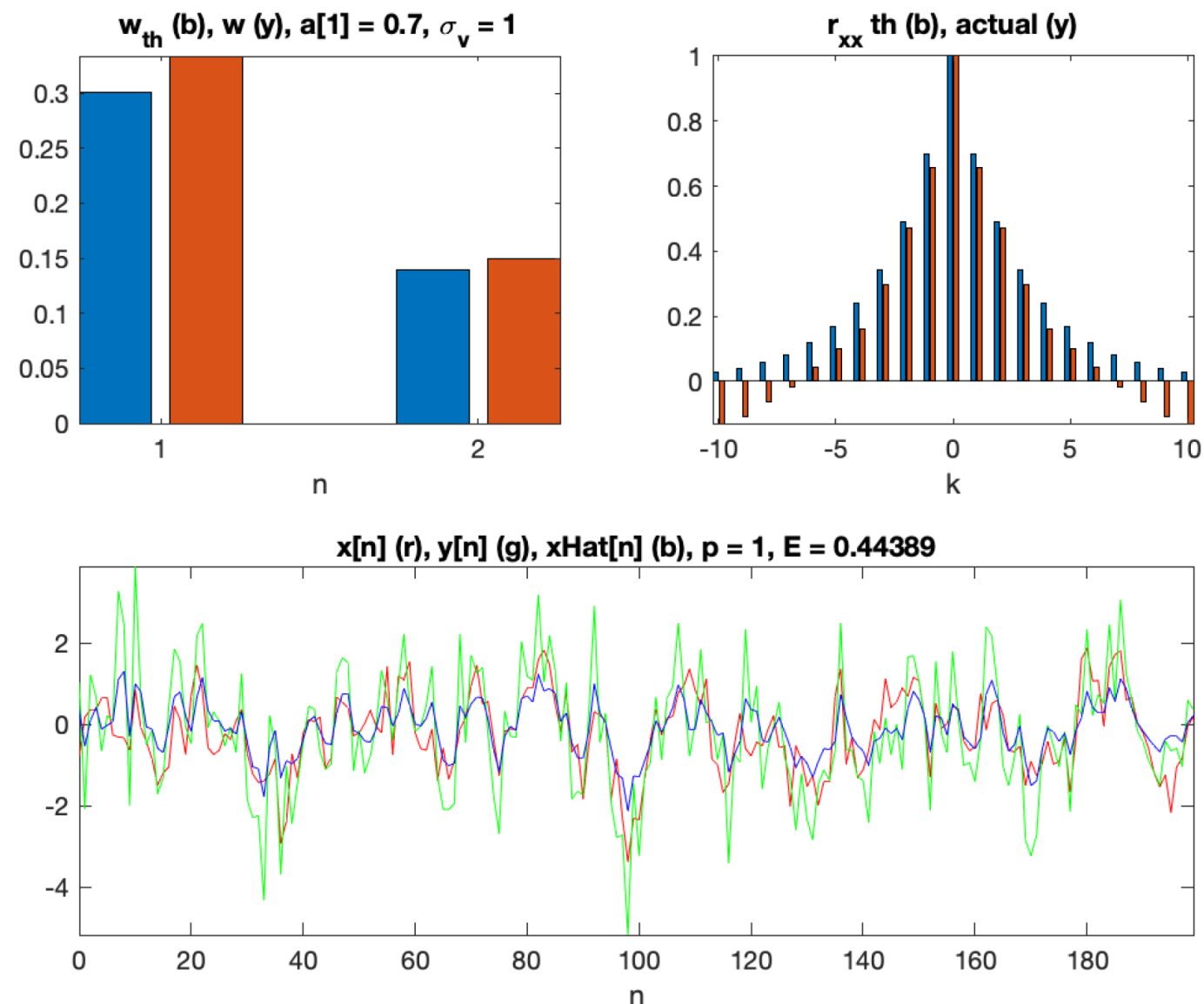
Example Linear Prediction (7.2.2)

- AR(1) process : $x[n] = -0.7x[n - 1] + w[n]$
- Measured **without noise** environment : $y[n] = x[n]$
- Wiener predict obtained (p=1) : $\mathbf{w} = (0.6117, 0.0704)$



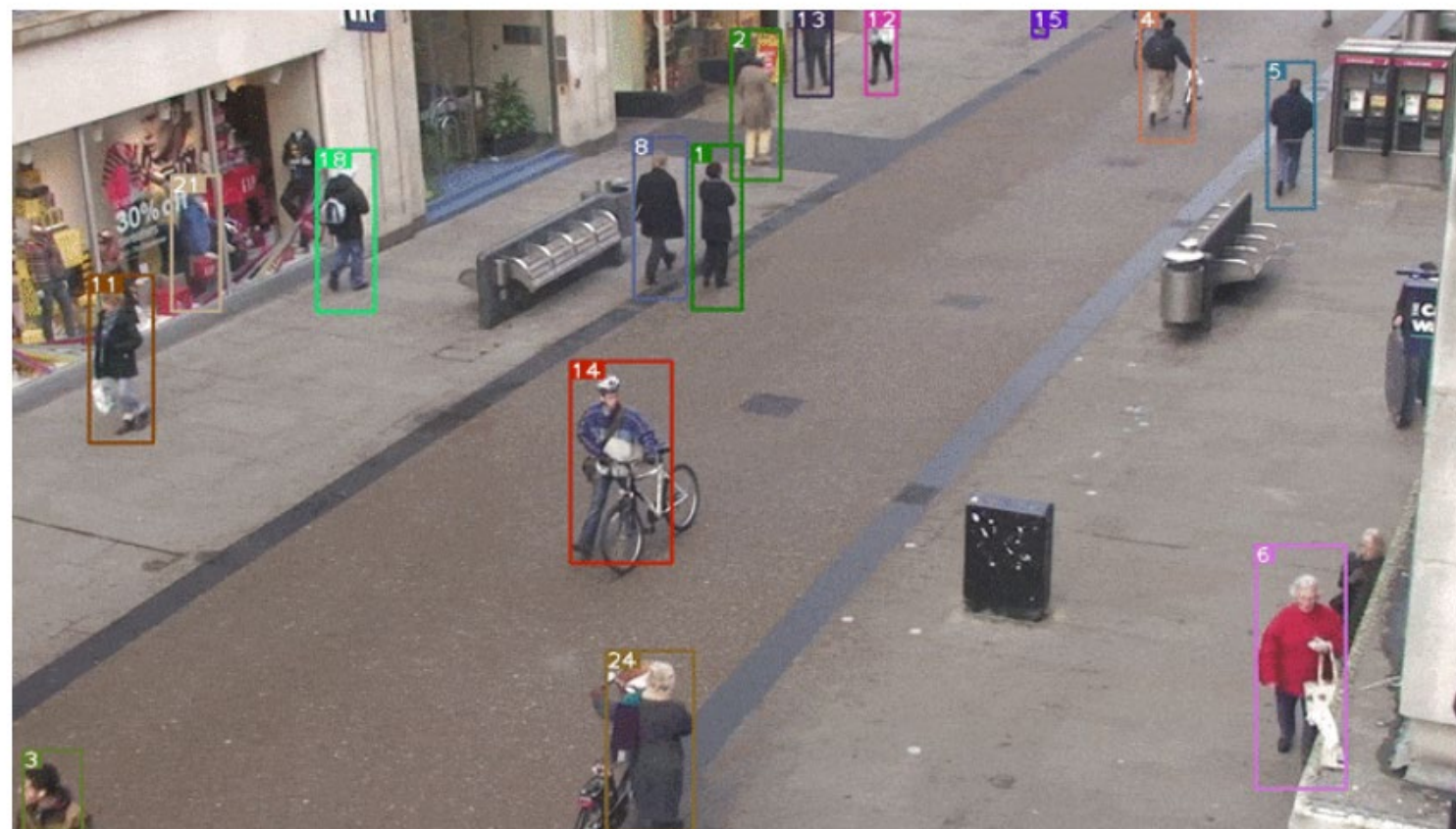
Example Linear Prediction (7.2.3)

- AR(1) process : $x[n] = -0.7x[n-1] + w[n]$
- Measured **with noise** environment : $y[n] = x[n] + v[n]$
- Wiener predict obtained (p=1) : $\mathbf{w} = (0.405, 0.238)$



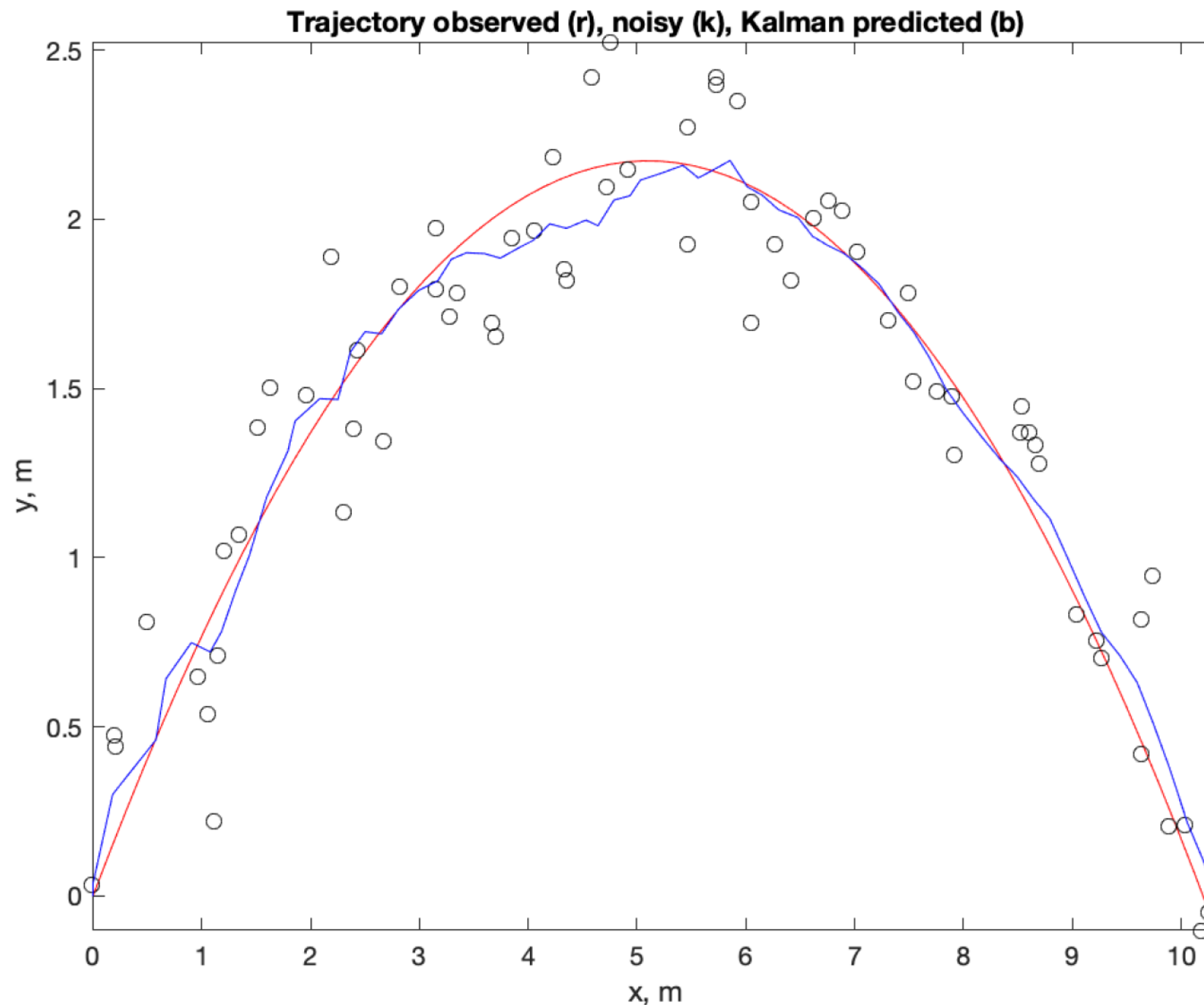
- FIR Wiener filter
- Kalman filter

- The goal of the *Kalman filter* is to estimate the unknown state vector of a dynamic system $X[n]$ based on the measurement $z[n]$ and others parameters.
- Kalman filter is heavily used in tracking problem, sometimes with Image processing. For example DeepSort tracking:



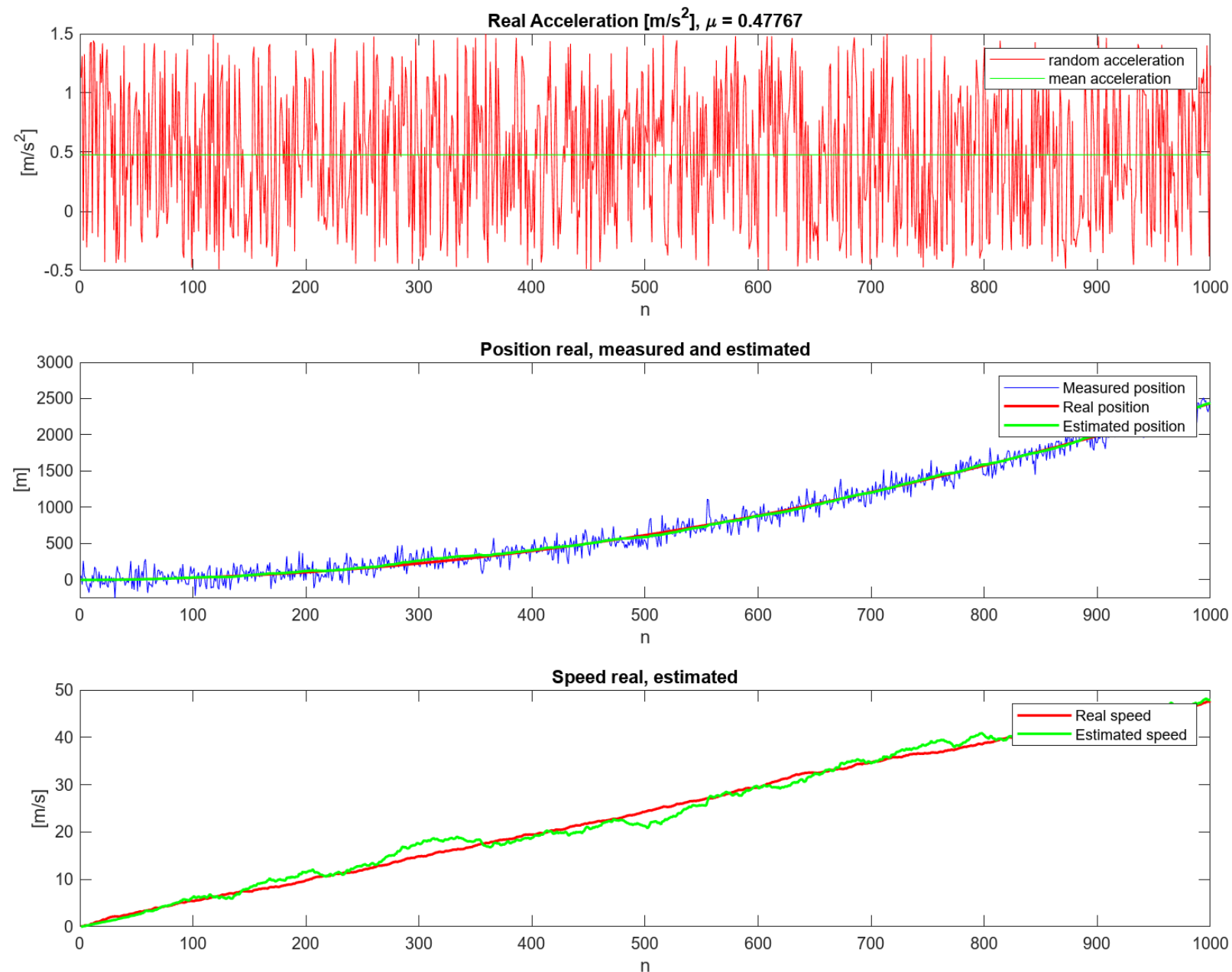
Example Kalman Body in gravitation

- Estimation of a noisy measurement

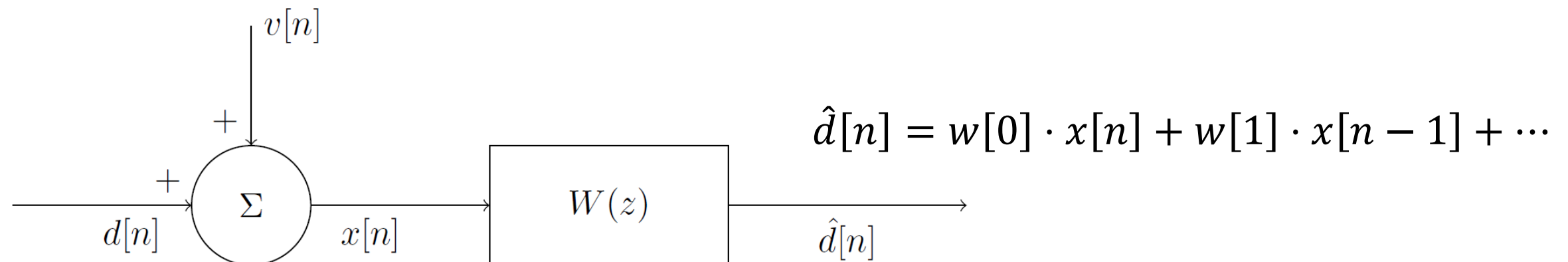


Example Kalman truck on rail

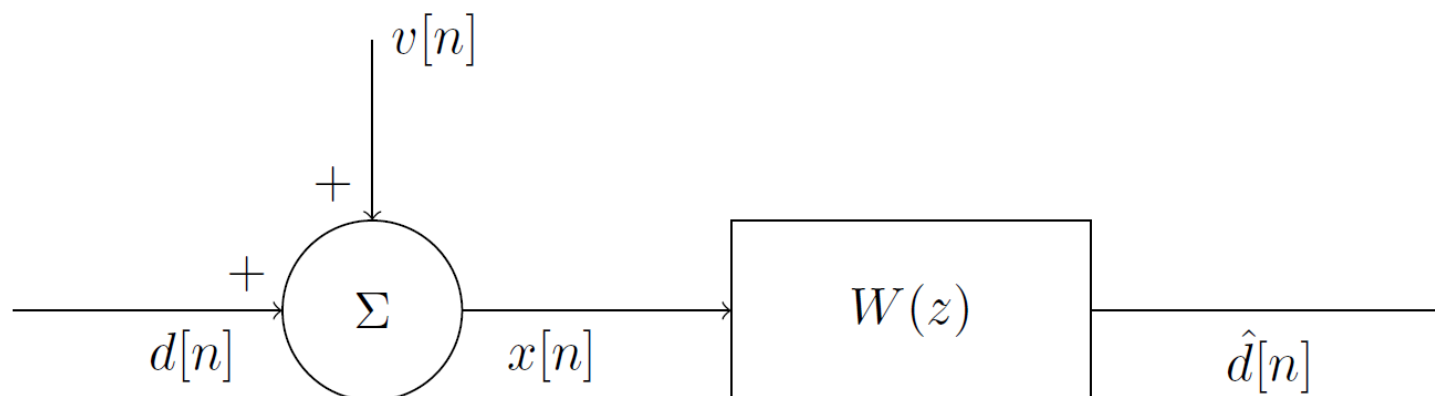
- Estimation of the position and speed of a truck with a noisy input and measurement



- With FIR Wiener, $d[n]$ can be estimated with a FIR filter

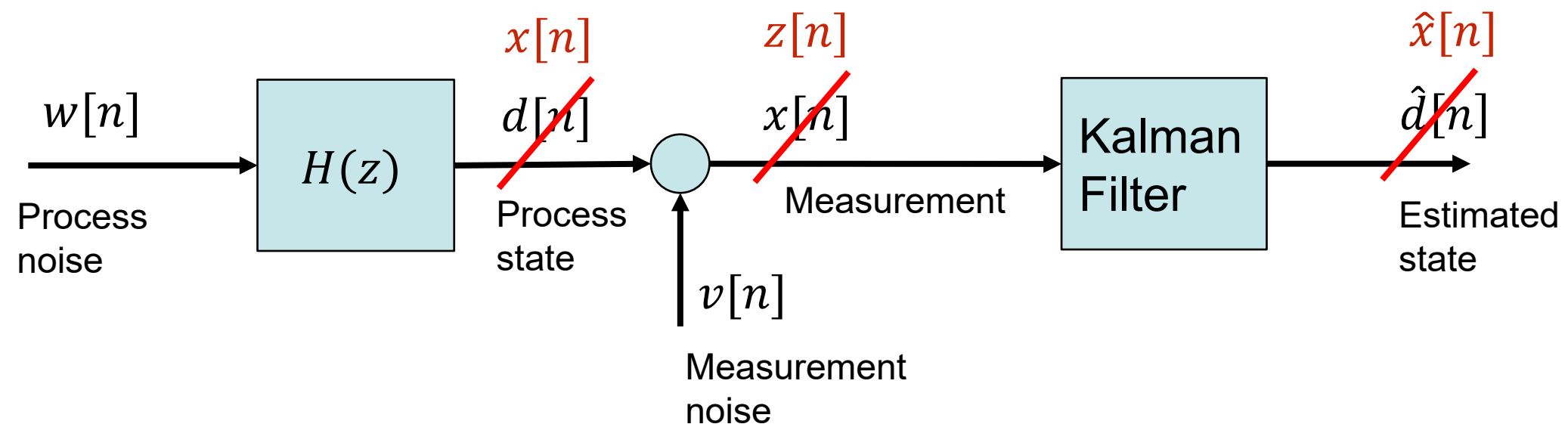


- With IIR Wiener, $d[n]$ can be estimated with a IIR filter



$$\begin{aligned} \hat{d}[n] &= w_b[0] \cdot x[n] + w_b[1] \cdot x[n - 1] + \dots + w_a[1] \cdot \hat{d}[n - 1] + w_a[2] \cdot \hat{d}[n - 2] + \dots \\ &= \sum_{k=0}^{q-1} w_b[k] x[n - k] + \sum_{k=1}^p w_a[k] \hat{d}[n - k] \end{aligned}$$

- **Wiener** is optimum if $x[n]$ and $d[n]$ are WSS.
- Another way to filter a signal is to **use the knowledge of the process** ($H(z)$) and the **knowledge of the measurement system**.
- This approach is known as **Kalman filter** and does not require that $x[n]$ and $d[n]$ are WSS.



- **Unfortunately, the notation used for Kalman is different in the literature!**

- Let's say that $x[n]$ is an AR(1) driven process ($w[n]$ and $v[n]$ are uncorrelated white noise process):

$$x[n] = a[1]x[n-1] + w[n] \quad \text{and} \quad z[n] = x[n] + v[n]$$

- Even if we know $a[1]$, it is not possible to predict $x[n]$ as we don't know $w[n]$
- For example, suppose we have :

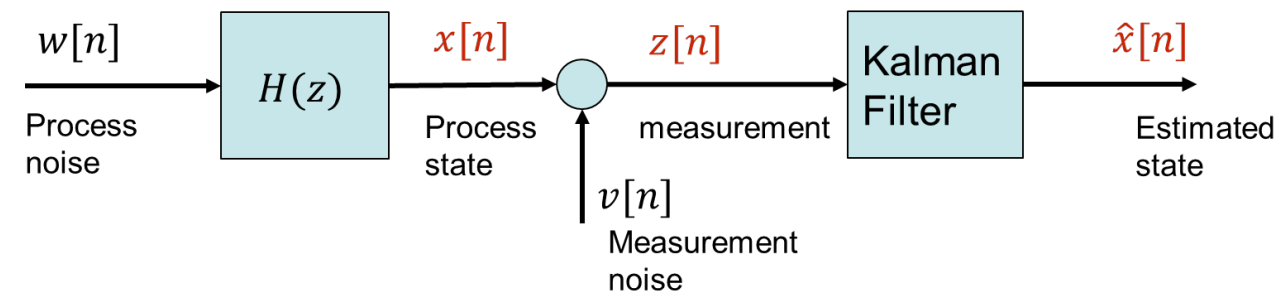
$$\hat{x}[n-1] = x[n-1]$$

- And you want to predict $x[n]$ by knowing $a[1]$:

$$\hat{x}[n] = a[1] \cdot \hat{x}[n-1]$$

- As $w[n]$ is a white noise, $x[n]$ cannot be estimated perfectly

- However, the measure $z[n]$ contains the information of $w[n]$ but unfortunately also a measurement noise $v[n]$



$$z[n] = a[1]x[n-1] + w[n] + v[n]$$

$$\alpha[n] = z[n] - a[1]\hat{x}[n-1] = w[n] + v[n]$$

Error (Innovation process) : new information brought by $z[n]$

- If the measurement noise is null $v[n] = 0$, the best estimated of $x[n]$ is :

$$\begin{aligned}\hat{x}[n] &= a[1] \cdot \hat{x}[n-1] + \alpha[n] \\ &= a[1] \cdot \hat{x}[n-1] + (z[n] - a[1]\hat{x}[n-1])\end{aligned}$$

- In reality, we don't know $a[1]$ and $v[n]$ is not null.
- To compensate, a Kalman gain **K** is added in factor to the new information brought by $z[n]$

$$\hat{x}[n] = \overbrace{a[1] \cdot \hat{x}[n-1]}^{\hat{z}[n]} + \underbrace{\mathbf{K}}_{\text{Kalman gain}} \cdot \underbrace{(z[n] - a[1]\hat{x}[n-1])}_{\substack{\text{Noisy} \\ \text{measurement}}}$$

$\hat{x}[n]$: Best estimate of $x[n]$ without noise
 $\mathbf{K}$: Kalman gain
 $z[n]$: Noisy measurement
 $\alpha[n] = z[n] - \hat{z}[n]$: Error (Innovation process)

- If $x[n]$ and $z[n]$ are not stationary, the equation is written :

$$\hat{x}[n] = a_{n-1}[1] \cdot \hat{x}[n-1] + \mathbf{K}[n] \cdot (z[n] - a_{n-1}[1]\hat{x}[n-1])$$

- If $x[n]$ is an AR(p) driven process, then it becomes:

$$x[n] = \sum_{k=1}^p a[k]x[n-k] + w[n] \quad \text{and} \quad z[n] = x[n] + v[n]$$

- Or in the matrix notation

$$\underbrace{\begin{bmatrix} x[n] \\ x[n-1] \\ x[n-2] \\ \vdots \\ x[n-p+1] \end{bmatrix}}_{\mathbf{x}[n]} = \underbrace{\begin{bmatrix} a[1] & a[2] & \dots & a[p] \\ 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{bmatrix}}_{\mathbf{F}} \underbrace{\begin{bmatrix} x[n-1] \\ x[n-2] \\ x[n-3] \\ \vdots \\ x[n-p] \end{bmatrix}}_{\mathbf{x}[n-1]} + \underbrace{\begin{bmatrix} w[n] \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}}_{\mathbf{w}[n]}$$

New state Transition matrix Previous state Noise process

$$\underbrace{z[n]}_{\text{Measurement}} = \underbrace{[1 \quad 0 \quad \dots \quad 0]}_{\text{Measurement matrix } \mathbf{H}} \underbrace{\begin{bmatrix} x[n] \\ x[n-1] \\ \vdots \\ x[n-p+1] \end{bmatrix}}_{\mathbf{x}[n]} + v[n]$$

Measurement matrix $\mathbf{H}$ New state Noise measurement

- Using the matrix, notation, we obtain:

The diagram illustrates the Kalman filter equations with labels for each term:

State Transition Equation:

$$\mathbf{x}[n] = \mathbf{F} \cdot \mathbf{x}[n-1] + \mathbf{w}[n]$$

Labels for the first equation:

- New state: $\mathbf{x}[n]$
- Transition matrix: $\mathbf{F}$
- Previous state: $\mathbf{x}[n-1]$
- Noise process: $\mathbf{w}[n]$

Measurement Equation:

$$\mathbf{z}[n] = \mathbf{H} \cdot \mathbf{x}[n] + \mathbf{v}[n]$$

Labels for the second equation:

- Measurement: $\mathbf{z}[n]$
- Measurement matrix: $\mathbf{H}$
- Noise disturbance: $\mathbf{v}[n]$

- So, the best estimate of $\mathbf{x}[n]$ is:

$$\hat{\mathbf{x}}[n] = \mathbf{F} \cdot \hat{\mathbf{x}}[n-1] + \mathbf{K}[n] [\mathbf{z}[n] - \mathbf{H} \cdot \mathbf{F} \cdot \hat{\mathbf{x}}[n-1]]$$

- A more heavy notation can be found

$$\hat{\mathbf{x}}[n] = \underbrace{\mathbf{F}}^{[px1] \quad [pxp] \quad [px1]} \cdot \underbrace{\hat{\mathbf{x}}[n-1]}^{[pxq] \quad [qx1] \quad [qx1]} + \mathbf{K}[n](\mathbf{z}[n] - \hat{\mathbf{z}}[n])$$

$$\hat{\mathbf{x}}[n|n] = \hat{\mathbf{x}}[n|n-1] + \mathbf{K}[n](\mathbf{z}[n] - \hat{\mathbf{z}}[n|n-1])$$

Estimation of $\mathbf{x}[n]$
knowing up to
sample n

Estimation of $\mathbf{x}[n]$
knowing up to
sample $n-1$

Innovation or
measurement
residual

$$\hat{\mathbf{z}}[n] = \mathbf{H} \cdot \underbrace{\mathbf{F} \cdot \hat{\mathbf{x}}[n-1]}^{[qx1] \quad [qxp] \quad [pxp] \quad [px1]}$$

$$\hat{\mathbf{z}}[n|n-1] = \mathbf{H} \cdot \hat{\mathbf{x}}[n|n-1]$$

Estimation of $\mathbf{z}[n]$
knowing sample
up to $n-1$

- Finally, the Kalman predictor is:

$$\begin{aligned}\hat{\mathbf{x}}[n|n] &= \hat{\mathbf{x}}[n|n-1] + \mathbf{K}[n][\mathbf{z}[n] - \hat{\mathbf{z}}[n|n-1]] \\ \hat{\mathbf{z}}[n|n-1] &= \mathbf{H} \cdot \hat{\mathbf{x}}[n|n-1]\end{aligned}$$

- With the corresponding error:

$$\begin{aligned}\overset{[p \times 1]}{\mathbf{e}}[n|n] &= \overset{[p \times 1]}{\mathbf{x}}[n] - \overset{[p \times 1]}{\hat{\mathbf{x}}}[n|n] \\ \mathbf{e}[n|n-1] &= \mathbf{x}[n] - \hat{\mathbf{x}}[n|n-1]\end{aligned}$$

- And the **covariance matrices errors** are:

$$\begin{aligned}\overset{[p \times p]}{\mathbf{P}}[n|n] &= E\{\overset{[p \times 1]}{\mathbf{e}}[n|n]\overset{[1 \times p]}{\mathbf{e}}^H[n|n]\} \\ \mathbf{P}[n|n-1] &= E\{\mathbf{e}[n|n-1]\mathbf{e}^H[n|n-1]\}\end{aligned}$$

- It can be shown that $\mathbf{P}[n|n-1]$ can be also obtained recursively :

$$\overset{[p \times p]}{\mathbf{P}}[n|n-1] = \overset{[p \times p]}{\mathbf{F}} \cdot \overset{[p \times p]}{\mathbf{P}}[n-1|n-1] \cdot \overset{[p \times p]}{\mathbf{F}'} + \overset{[p \times p]}{\mathbf{Q}_w}$$

- The Kalman gain is obtained by minimizing the quadratic error :

$$\xi[n] = \text{tr}\{\mathbf{P}[n|n]\} = E\{\|\mathbf{e}[n|n]\|^2\} \Rightarrow \frac{\partial \xi[n]}{\partial \mathbf{K}} = 0$$

$$\Rightarrow \overset{[p \times p]}{\mathbf{K}}[n] = \overset{[p \times p]}{\mathbf{P}}[n|n-1] \overset{[p \times q]}{\mathbf{H}}^H \left[\overset{[q \times p]}{\mathbf{H}} \cdot \overset{[p \times p]}{\mathbf{P}}[n|n-1] \cdot \overset{[p \times q]}{\mathbf{H}}^H + \overset{[q \times q]}{\mathbf{Q}_v}[n] \right]^{-1}$$

- The covariance of the error $\mathbf{e}[n|n]$ can be rewritten :

$$\mathbf{P}[n|n] = [\mathbf{I} - \mathbf{K}[n]\mathbf{H}] \cdot \mathbf{P}[n|n-1]$$

Where $\mathbf{Q}_v[n]$ is the covariance matrix of the measurement noise $\mathbf{v}[n]$ and $\mathbf{Q}_w[n]$ is the covariance matrix of the process noise

- To summarize, the implementation of the Kalman filter is recursive :

For $n=1,2,\dots$ compute :

$$\text{Prediction} \left\{ \begin{array}{l} \hat{\mathbf{x}}[n|n-1] = \mathbf{F} \cdot \hat{\mathbf{x}}[n-1|n-1] \\ \mathbf{P}[n|n-1] = \mathbf{F} \cdot \mathbf{P}[n-1|n-1] \cdot \mathbf{F}^H + \mathbf{Q}_w[n] \end{array} \right.$$

$$\text{Update} \left\{ \begin{array}{l} \mathbf{K}[n] = \mathbf{P}[n|n-1] \mathbf{H}^H [\mathbf{H} \cdot \mathbf{P}[n|n-1] \cdot \mathbf{H}^H + \mathbf{Q}_v[n]]^{-1} \\ \hat{\mathbf{x}}[n|n] = \hat{\mathbf{x}}[n|n-1] + \mathbf{K}[n] [\mathbf{z}[n] - \mathbf{H} \cdot \hat{\mathbf{x}}[n|n-1]] \\ \mathbf{P}[n|n] = [\mathbf{I} - \mathbf{K}[n] \mathbf{H}] \cdot \mathbf{P}[n|n-1] \end{array} \right.$$

- All that needs to be done to complete the recursion is to initialize at time $n=0$. The initial estimate is very often chosen

$$\hat{\mathbf{x}}[0|0] = E\{\mathbf{x}(0)\}$$

$$\mathbf{P}[0|0] = E\{\mathbf{x}(0)\mathbf{x}^H(0)\}$$

- **Important observation:**
 - $\mathbf{K}[n]$ and $\mathbf{P}[n|n]$ do not depend of $\mathbf{x}[n]$. Therefore, they can be computed off-line prior to any new measurement.